Hidden Creators

An Expected Threat (xT) player impact report: finding players whose goals and assists understate their contribution

Jun Jeon · September 2026 · Data: StatsBomb Open Data 2015/16 · La Liga, Premier League, Serie A, Ligue 1 · 1,517 matches

Key findings

- Goals and assists do not capture players who carry or pass the ball into dangerous areas. I split the pitch into a 12×8 grid, estimated for each cell the probability that a possession there ends in a goal (xT), and credited every completed pass and carry with the xT it added.
- The grid correlates at 0.995 with Karun Singh's published reference grid and at 0.996 with an empirical grid counted directly from real possession outcomes. Ranking players with the empirical grid instead gives a rank correlation of 0.984, so the conclusions do not hinge on the model's assumptions.
- 36 attacking players sit in the top 20% of their position for xT but at or below the position median for goals + assists. The **26** of them at clubs finishing 7th or lower form the main shortlist. Wilfried Zaha (Crystal Palace) produced xA 4.0 but 1 assist; Luca Siligardi (Hellas Verona) produced xA 3.7 and 0 assists.
- For centre-backs and defensive midfielders, where goals + assists is not a meaningful benchmark, a separate ranking of 74 ball-progressing defenders is provided.

1. The problem

The attacking metrics most used to judge players (goals, assists and xG) all look at the **last two actions** of an attack: the shot and the pass before it. A 30-yard carry from the halfway line, or a line-breaking pass followed by one more pass from a teammate, leaves no trace in any of them.

Expected Threat (xT) fills that gap. It assigns every zone of the pitch the probability that having the ball there eventually leads to a goal, and credits any action that moves the ball from a lower-value zone to a higher-value one with the difference. The goal of this report is to measure each player's ball progression with xT and to turn it into a shortlist a scout can act on: **players whose goals and assists understate what they contribute**.

The project started as a model predicting whether transfers between leagues would succeed. I dropped it: future performance cannot be validated with open data, and the effect of a move cannot be separated from managerial changes, tactical fit or injuries. xT evaluates events that have already happened, so it can be checked against a public reference and against real outcomes.

2. Data and scope

The analysis uses the 2015/16 season of StatsBomb Open Data, chosen because it is the only season released for all of the big five leagues. On inspection, however, the Bundesliga release contains only Bayer Leverkusen's 34 matches (every other team appears in just its two games against Leverkusen), so it was excluded. The Champions League release is the final only, which falls outside a league-season analysis. The final scope is four leagues.

League	Matches	Passes	Carries	Shots
La Liga	380	365,288	276,171	9,168
Premier League	380	368,619	276,949	9,908
Serie A	380	373,667	280,820	9,998
Ligue 1	377	374,239	283,256	8,814

The Ligue 1 release contains 377 of 380 matches. In the league tables this leaves Paris Saint-Germain three points short of their actual total, but the gap between 6th and 7th is large enough that the top-6 split is unaffected.

2.1 Data-quality checks

All 1,552 downloaded match files were parsed in full, not sampled, and reconciled against known external facts.

Check	Result
Goals counted from events (shot goals + own goals) vs official scores	All 1,552 matches match, down to home/away team level
Minutes played	No duplicates or negative values. The 157 team-matches below 11 × match length are all explained by red cards
Attacking-direction normalisation	No matches with a flipped direction
Off-pitch coordinates	28 cases, at most 0.9 yards → clipped to the pitch edge
xA linkage (shot → key pass)	All 27,035 shots with a key pass link to a pass by the same team; goals from key passes match the 2,596 flagged assists exactly
League tables computed from results	Match actual totals, e.g. Leicester 81 pts, Barcelona 91, Juventus 91

3. Method

The xT of each zone z is found by iterating the equation below. From zone z the player either shoots (probability s) or moves the ball to another zone (probability m). A shot is worth the zone's scoring probability g; a move inherits the xT of the zone where the ball arrives. A failed move is worth zero.

$$xT(z) = s(z) \times g(z) + m(z) \times \sum_{z} T(z \to z) \times xT(z)$$

T(z→z) is the share of moves from z that successfully arrive in z. A player's contribution is the sum of xT(end zone) − xT(start zone) over their completed actions, expressed per 90 minutes. Moving the ball backwards is negative.

3.1 Design decisions

Decision	Reason
12×8 grid (each cell 10×10 yards)	Same resolution as the public reference grid, so the two can be compared directly
Open-play passes and carries only (no set pieces)	The subject is open-play creation; keeps corner and free-kick locations from inflating zone values
Carries under 3 yards excluded	StatsBomb logs a carry between receiving and passing even without movement, double-counting the zone (510,615 carries)
Pass outcomes 'Unknown' and 'Injury Clearance' excluded	Neither success nor failure; kicking the ball out for an injured player is not a failed pass (12,013 passes)
Open-play shots only (no penalties)	Penalties (75% conversion) would inflate the value of the zones inside the box
Penalty shoot-outs excluded	Not part of open play
Players identified by ID	9 players appear under two name spellings; 115 changed club during the season

After filtering, the model uses 1,916,597 moves and 35,732 open-play shots. The code checks during the calculation that probabilities sum to one and values stay within 0–1, and stops if they do not.

4. Validating the model

Value iteration converged after 42 iterations, with the largest change in any cell below 0.000001. The grid was then checked against three benchmarks.

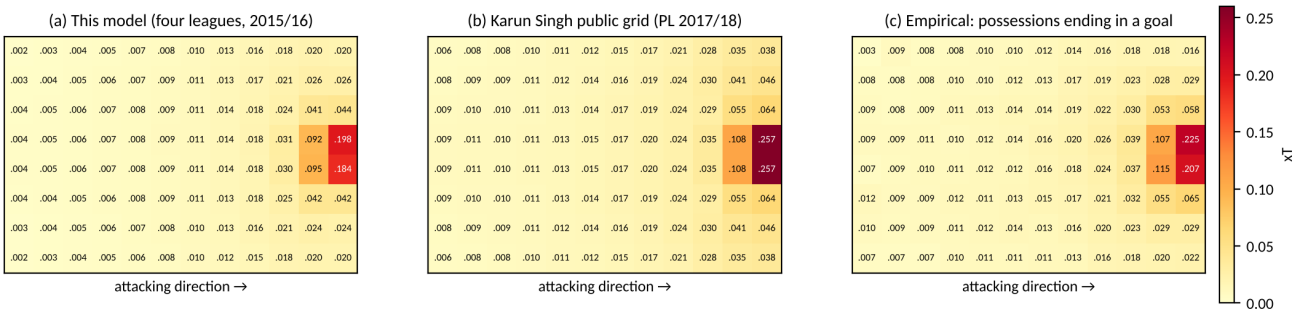


Figure 1. xT by zone. (a) This model. (b) Karun Singh's published grid (Premier League 2017/18). (c) Empirical values: the share of actions in each zone whose possession ended in a goal.

Comparison	Pearson	Spearman	Interpretation
This model vs public grid	0.995	0.994	Same shape; values about 40% lower (see 4.1)
This model vs empirical	0.996	0.967	85–90% of empirical near the box; about half in the own half
Each league on its own vs pooled	≥ 0.996	≥ 0.999	No single league drives the result

4.1 Tracing the magnitude gap

The shape matched, but the values were lower than the public grid (around halfway: 0.010 vs 0.016). Rather than accept a high correlation and move on, I tested each possible explanation.

Hypothesis	Test	Result
My design choices (carry threshold, no set pieces)	Refit under alternative settings	Rejected. Ratio stays at 0.62–0.66; removing carries entirely moves it further away (0.36)
Failed passes are often won back by the same team	Check who has the ball after a failed pass	Partial. Only 7% in the own third, so not the main cause
Set pieces within a possession are invisible to the model	Look for set pieces inside scoring possessions	Rejected. StatsBomb starts a new possession at every set piece (0 cases)
Pooling strong and weak teams lowers the averages	Refit within team-quality quartiles	Rejected. The gap is unchanged within each quartile
The Markov (memoryless) assumption	Compare model vs empirical probability of reaching the final third, with no scoring involved	**Confirmed.** The model also understates progression, more so the further from goal (ratio 0.68 → 0.95)

xT only knows which zone the ball is in, not the flow of the possession (a settled build-up versus a scramble). It therefore understates how often attacks that must pass through many steps, such as those starting in a team's own half, keep going. This is a known property of the method, not a calculation error. The public grid was built from a different provider's data and season, with different event definitions, so its absolute values need not match.

4.2 Does this limitation change the conclusions?

I recomputed every player's contribution using the empirical grid instead of the model grid. Across the 1,358 players with 900+ minutes the rank correlation is 0.984; 18 of the top 20, 45 of the top 50 and 91 of the top 100 are the same. Own-half zones have very small absolute values, so they barely separate players.

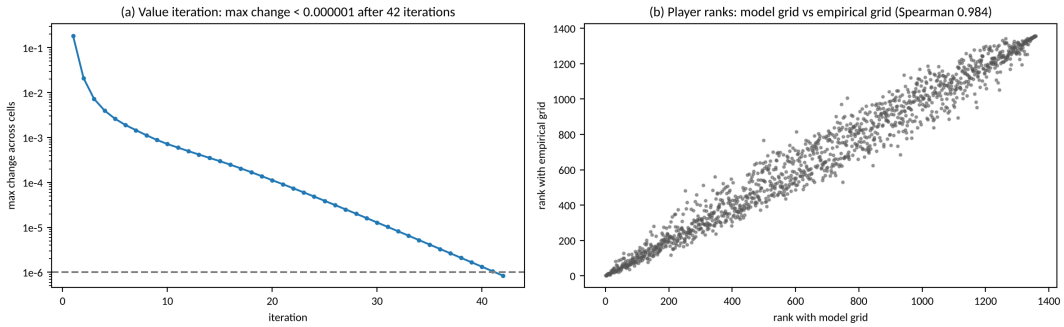


Figure 2. (a) Convergence of value iteration. (b) Player ranks under the model grid vs the empirical grid.

5. Player xT

1,358 players played 900+ minutes in the four leagues. The top ten by xT per 90 are below: Messi, Neymar, Di María and Özil, all known for progressive passing and dribbling, lead the list.

#	Player	Club	Position	Mins	xT/90	Pass	Carry	npG/90	A/90
1	Lionel Messi	Barcelona	Winger	2,807	0.425	0.251	0.174	0.74	0.48
2	Neymar	Barcelona	Winger	3,139	0.415	0.184	0.231	0.54	0.34
3	Ángel Di María	Paris Saint-Germain	Winger	2,081	0.393	0.295	0.098	0.43	0.74
4	Gerard Deulofeu	Everton	Winger	1,447	0.382	0.229	0.153	0.12	0.56
5	Jefferson Montero	Swansea City	Winger	1,415	0.334	0.193	0.141	0.00	0.13
6	Burgui	Espanyol	Winger	1,043	0.328	0.182	0.146	0.09	0.26
7	Mesut Özil	Arsenal	Attacking mid	3,135	0.324	0.274	0.050	0.17	0.55
8	Dries Mertens	Napoli	Winger	1,192	0.320	0.166	0.154	0.38	0.38
9	Dušan Tadić	Southampton	Winger	2,348	0.318	0.251	0.067	0.23	0.42
10	James Rodríguez	Real Madrid	Central mid	1,560	0.317	0.268	0.049	0.40	0.46

Pass and Carry split xT/90 by action type. npG/90 = non-penalty goals per 90; A/90 = assists per 90.

5.1 Position effects

Median xT per 90 by position: winger 0.157, wide mid 0.152, attacking mid 0.130, full-back 0.122, central mid 0.102, defensive mid 0.073, centre-back 0.055, forward 0.048. Forwards ranking below centre-backs is not an error. 41% of forwards' completed actions reduce xT, because much of their work is receiving near the box and laying the ball back. xT credits the player who **sends** the ball into a dangerous zone, not the one who receives it there or shoots. Even Ibrahimović, Suárez and Benzema, all scoring more than a goal per game, sit at 0.12–0.17. Players must therefore be compared **within their position**.

5.2 Minimum minutes

xT per 90 is noisy for players with few minutes. I split each player's matches into odd and even halves and measured how well the two halves agree within position.

Minutes	Players	Split-half correlation	Full-season reliability
900–1,350	269	0.54	0.70
1,350–1,800	236	0.67	0.81
1,800–2,400	338	0.65	0.79
2,400–4,000	415	0.70	0.82

Full-season reliability applies the Spearman–Brown correction. Reliability settles at about 0.8 from 1,350 minutes (15 full matches), so the overall table uses 900 minutes and the shortlist uses 1,350.

6. Shortlist: undervalued creators

The pool is attacking players (full-backs; central, wide and attacking midfielders; wingers; forwards) with 1,350+ minutes. A player is flagged when both conditions hold:

(1) xT per 90 in the top 20% of the position (2) non-penalty goals + assists per 90 at or below the position median

Of the 36 players flagged, the 10 at top-6 clubs, whose value is more likely to be widely known, were separated out. The remaining 26 form the main shortlist.

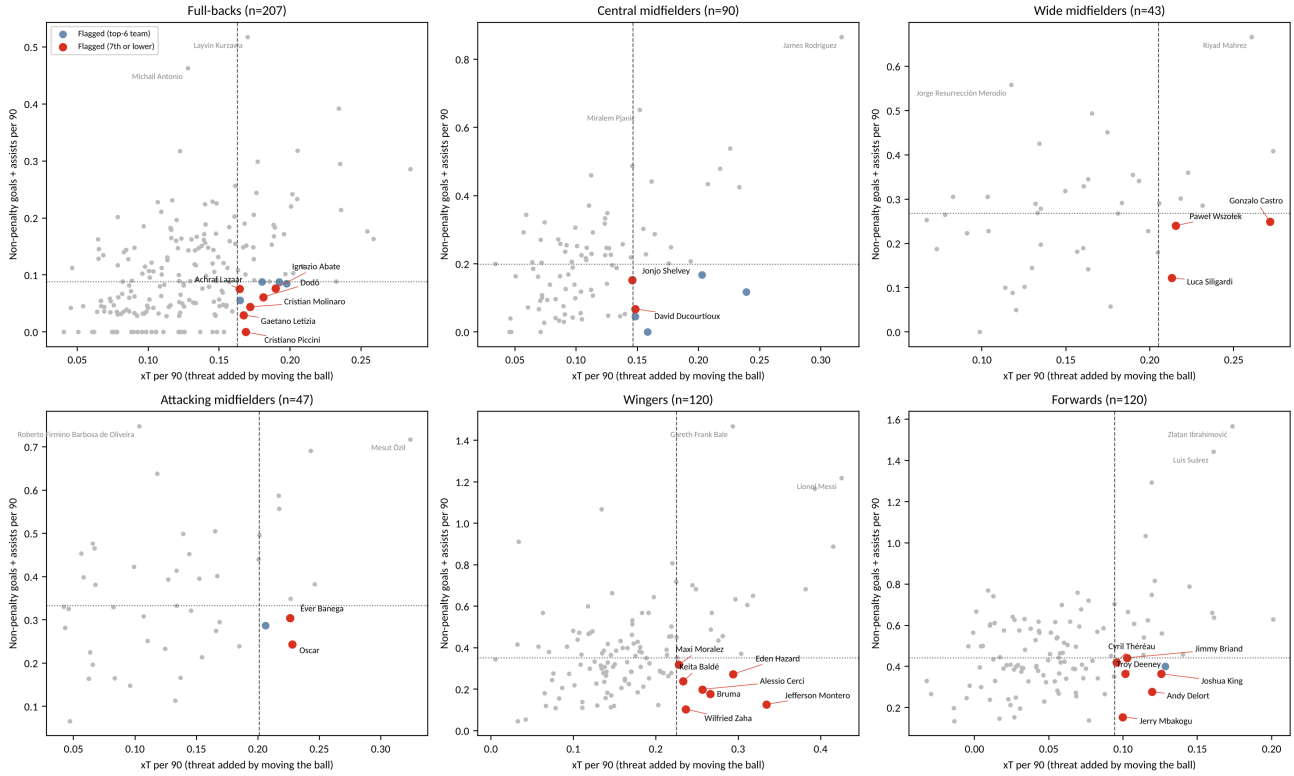


Figure 3. xT per 90 against non-penalty goals + assists per 90, by position. The dashed vertical line marks the top 20% for xT; the dotted horizontal line marks the goals + assists median. Red: shortlist (clubs 7th or lower). Blue: flagged players at top-6 clubs. Grey labels: the two highest goals + assists players in each position, for reference.

6.1 Main shortlist (26 players)

Player	Club (finish)	Position	Mins	xT/90	Within pos.	G+A/90 (median)	xA / A	Carry share
Gonzalo Castro	Málaga (8th)	Wide mid	1,447	0.272	top 2%	0.25 (0.27)	3.9 / 3	26%
Jefferson Montero	Swansea City (12th)	Winger	1,415	0.334	top 3%	0.13 (0.35)	2.7 / 2	42%
Oscar	Chelsea (10th)	Attacking mid	1,847	0.228	top 6%	0.24 (0.33)	3.0 / 3	32%
Eden Hazard	Chelsea (10th)	Winger	2,312	0.294	top 8%	0.27 (0.35)	4.5 / 3	41%
Joshua King	AFC Bournemouth (16th)	Forward	1,976	0.126	top 8%	0.36 (0.44)	2.7 / 2	62%
Ignazio Abate	AC Milan (7th)	Full-back	2,346	0.190	top 8%	0.08 (0.09)	1.9 / 1	19%
Andy Delort	Stade Malherbe Caen (7th)	Forward	3,246	0.120	top 9%	0.28 (0.44)	4.1 / 2	52%
Éver Banega	Sevilla (7th)	Attacking mid	1,773	0.226	top 11%	0.30 (0.33)	4.4 / 2	24%
Dodô	Sampdoria (15th)	Full-back	1,470	0.181	top 11%	0.06 (0.09)	1.8 / 1	24%
Bruma	Real Sociedad (9th)	Winger	1,526	0.266	top 12%	0.18 (0.35)	3.2 / 1	52%
Alessio Cerci	AC Milan (7th)	Winger	1,353	0.257	top 13%	0.20 (0.35)	1.7 / 2	39%
Paweł Wszołek	Hellas Verona (20th)	Wide mid	1,875	0.215	top 14%	0.24 (0.27)	4.6 / 5	33%
Cristian Molinaro	Torino (12th)	Full-back	2,039	0.172	top 15%	0.04 (0.09)	2.0 / 0	33%
Wilfried Zaha	Crystal Palace (15th)	Winger	2,592	0.236	top 16%	0.10 (0.35)	4.0 / 1	46%

Player	Club (finish)	Position	Mins	xT/90	Within pos.	G+A/90 (median)	xA / A	Carry share
Luca Siligardi	Hellas Verona (20th)	Wide mid	1,468	0.213	top 16%	0.12 (0.27)	3.7 / 0	27%
Cristiano Piccini	Real Betis (10th)	Full-back	1,472	0.169	top 16%	0.00 (0.09)	1.8 / 0	29%
Keita Baldé	Lazio (8th)	Winger	1,885	0.233	top 17%	0.24 (0.35)	3.9 / 1	72%
Jimmy Briand	Guingamp (16th)	Forward	3,056	0.103	top 17%	0.44 (0.44)	5.7 / 8	45%
Maxi Moralez	Atalanta (13th)	Winger	1,414	0.228	top 18%	0.32 (0.35)	3.0 / 4	16%
Troy Deeney	Watford (13th)	Forward	3,450	0.101	top 18%	0.37 (0.44)	8.2 / 7	16%
David Ducourtioux	Gazélec Ajaccio (19th)	Central mid	2,665	0.148	top 18%	0.07 (0.20)	1.8 / 2	10%
Gaetano Letizia	Carpi (18th)	Full-back	3,026	0.167	top 18%	0.03 (0.09)	2.7 / 0	26%
Jerry Mbakogu	Carpi (18th)	Forward	1,750	0.100	top 18%	0.15 (0.44)	3.2 / 2	53%
Cyril Théréau	Udinese (16th)	Forward	2,997	0.096	top 19%	0.42 (0.44)	4.9 / 3	51%
Achraf Lazaar	Palermo (17th)	Full-back	2,378	0.164	top 19%	0.08 (0.09)	1.3 / 1	19%
Jonjo Shelvey	Swansea City (12th)	Central mid	2,366	0.146	top 20%	0.15 (0.20)	4.4 / 4	22%

Club finish is the final league position, computed from match results. The bracket in G+A/90 is the position median. Carry share is the proportion of the player's xT generated by carries.

6.2 Case studies

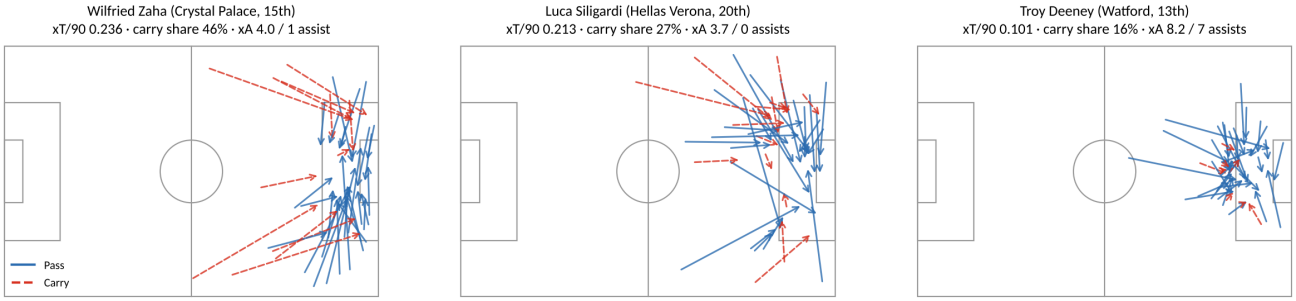


Figure 4. Each player's 40 actions with the largest xT gain. Solid blue: passes. Dashed red: carries. Attacking left to right.

Wilfried Zaha: 46% of his xT came from carries, and xA 4.0 turned into 1 assist. Unrecorded ball-carrying and chances lost to teammates' finishing combine here. In the matches he played he generated 20% of Crystal Palace's xT, the most in the squad, for a club that finished 15th.

Luca Siligardi: xA 3.7 and 0 assists, the clearest case on the list of chances created with nothing to show for it. The pitch map shows mostly crosses and cut-backs from the flank into the box. Playing for the club that finished 20th, poor finishing around him likely erased his record.

Troy Deeney: xA 8.2, 3rd among forwards with 1,350+ minutes behind Luis Suárez and Paulo Dybala, with 84% of his xT coming from passes. A striker usually judged on goals, his role was to a large extent linking play in front of the box.

Elsewhere, Keita Baldé (Lazio) has the highest carry share on the list at 72%, and Jefferson Montero (Swansea City) ranks in the top 3% of wingers for xT while his goals + assists per 90 (0.13) was less than half the position median (0.35).

6.3 Reading the shortlist

These criteria find players **whose goals and assists understate their contribution**, not players who are cheap in the market; final league position is only a proxy for market status. Chelsea, for example, finished 10th this season, which puts Eden Hazard and Oscar on the shortlist. Hazard's 2015/16 is often cited as the worst season of his career on output, yet his xT was in the top 8% of wingers and 41% of it came from carries, which suggests the dip in output was not a dip in contribution.

The top-6 reference list (Andrés Iniesta, Stevan Jovetić, Šime Vrsaljko, Sergi Darder, Jérémy Pied, Cédric Soares, Michael Carrick, Mathieu Valbuena, Alex Telles, Florent Balmont) is dominated by players whose value was already recognised, such as Iniesta and Carrick. It is more useful as a sanity check that the metric agrees with established opinion than as a scouting list.

7. Ball-progressing defenders

The median goals + assists per 90 for centre-backs is 0.05, so condition (2) holds for almost every one of them. Centre-backs and defensive midfielders were therefore ranked separately, without the goals + assists condition, taking the top 20% for xT within position (74 players). Mascherano, Bonucci, Thiago Silva and Kroos, all known for their passing out from the back, rank near the top, which indicates that xT captures defenders' progression. Below are the top 12 at clubs finishing 7th or lower.

Player	Club (finish)	Position	Mins	xT/90	Within pos.	Pass share	Move success
Cesc Fàbregas	Chelsea (10th)	Defensive mid	3,037	0.254	top 1%	80%	88%
Nikola Maksimović	Torino (12th)	Centre-back	1,508	0.155	top 1%	70%	89%
Cesare Bovo	Torino (12th)	Centre-back	1,874	0.132	top 1%	82%	88%
Gareth Barry	Everton (11th)	Defensive mid	2,971	0.167	top 2%	86%	87%
Tomás Rincón	Genoa (10th)	Defensive mid	3,049	0.165	top 3%	74%	90%
Étienne Didot	Toulouse (17th)	Defensive mid	1,660	0.163	top 3%	92%	78%
Walid Mesloub	Lorient (15th)	Defensive mid	2,200	0.161	top 4%	78%	87%
Dani Ceballos	Real Betis (10th)	Defensive mid	2,202	0.152	top 5%	69%	88%
Francesco Lodi	Udinese (16th)	Defensive mid	1,945	0.143	top 6%	86%	86%
Sylvain Armand	Rennes (8th)	Centre-back	3,116	0.105	top 6%	86%	88%
Hilton	Montpellier (11th)	Centre-back	3,295	0.105	top 7%	87%	82%
Emiliano Moretti	Torino (12th)	Centre-back	3,308	0.104	top 7%	78%	91%

Maksimović (Torino), Rincón (Genoa) and Ceballos (Real Betis) moved to Napoli, Juventus and Real Madrid within the following two years. These examples were picked with hindsight, so they do not validate the metric's predictive power and are noted only as a point of interest.

8. Limitations and next steps

Markov assumption. As shown in section 4, xT ignores the flow of a possession. Own-half values are understated, but player rankings are barely affected (rank correlation 0.984). Adding possession context, such as counter-attack status or pressing intensity, to the state is a possible extension.

No cost for losing the ball. Following the original xT formulation, actions that lose possession score zero. Players who attempt many risky passes and often lose the ball may be favoured. Charging lost possessions with the opponent's resulting xT is the next step.

Team context. Players in high-possession teams see more of the ball and accumulate xT more easily. Expressing each player's xT as a share of the team total is one way to adjust for this.

Off-ball movement. Event data records only on-ball actions. A player who runs into space and receives a pass has that value credited to the passer.

Scope. One season, four leagues, 2015/16. Without transfer fees, ages or contract data, 'undervalued in the market' is approximated only by league position.

Appendix: reproducing the results

Every result can be reproduced from scratch with a single script. Running **bash run_all.sh** downloads the StatsBomb data (about 4.4 GB, roughly a minute) and then runs data validation, the model, player aggregation, the figures and this PDF in order. Each step contains automatic checks and stops if a value is inconsistent. Running it from an empty folder reproduced every extracted table, grid, ranking, shortlist and the text of this report exactly.

Step	Script	Purpose
1	extract_events.py	Parse all 1,552 matches into analysis tables + data-quality counters
2	validate_data.py	Reconcile goals, minutes and coordinates with external ground truth
3	build_xt_model.py	Fit the xT grid; compare with the public grid and across leagues
4	xt_empirical_check.py	Empirical grid; tests of the magnitude-gap hypotheses
5	xt_diagnostics.py	Sensitivity of the grid to design choices
6	build_standings.py	Final league tables from match results
7	build_player_xt.py	Player xT, xA, goals and assists; model vs empirical ranking check
8	reliability_check.py	Split-half reliability by minutes played
9	flag_undervalued.py	Shortlist and ball-progressing defender rankings
10	make_figures.py / build_report.py	Figures and this PDF

Data: StatsBomb Open Data (github.com/statsbomb/open-data). Reference grid: Karun Singh, 'Introducing Expected Threat (xT)', karun.in/blog/expected-threat.html.